Journal of Cataract & Refractive Surgery

Accuracy of the ESCRS Toric IOL Power Calculator: Analysis of 6,753 Eyes

--Manuscript Draft--

Manuscript Number:	
Article Type:	Full Length Article
Section/Category:	Cataract
Full Title:	Accuracy of the ESCRS Toric IOL Power Calculator: Analysis of 6,753 Eyes
Corresponding Author:	Li Wang, MD, PhD Baylor College of Medicine Houston, Texas UNITED STATES
First Author:	Janusz Skrzypecki, Ph.D., M.D.
Order of Authors:	Janusz Skrzypecki, Ph.D., M.D.
	Douglas Koch
	Li Wang
Abstract:	Purpose To compare the predictive accuracy of four toric IOL formulas (Barrett, EVO, Hoffer® QST, Kane) integrated in the ESCRS calculator. Barrett and EVO were tested with measured (Barrett M and EVO M) and predicted (Barrett P and EVO P) posterior corneal curvature. Design retrospective data analysis Setting multiple surgical centers in the USA Methods We analyzed 6753 eyes implanted with non-toric IOLs. Preoperative biometry, including posterior corneal curvature, was obtained using the IOLMaster 700. Predicted residual astigmatism was calculated via the proprietary websites of the four toric formulas integrated into the ESCRS IOL Power Calculator. Primary endpoints were the mean vector magnitude error and proportions of eyes with vector magnitude prediction errors ≤0.50 D. Results Kane demonstrated the lowest mean vector magnitude prediction error (0.537 D), significantly lower than EVO P (0.561 D), EVO M (0.57 D), Barrett M (0.579 D), Barrett P (0.583 D), and Hoffer®QST (0.622 D) (all $p<0.001$). This result was consistent in with-the-rule, against-the-rule, and oblique subgroups. Kane also achieved the highest proportion of eyes with prediction error ≤0.50 D ($p<0.001$). For all formulas, centroid values were significantly different from zero. The Kane formula showed the lowest total variances, significantly lower than all other formulas ($p<0.001$). Conclusions The Kane formula showed statistically superior predictive accuracy with the lowest errors and variances and the highest proportion of eyes within 0.50 D. However, differences among formulas were small, and their clinical relevance for toric IOL outcomes remains uncertain.
Keywords:	IOL power; astigmatism; ESCRS calculator

Powered by Editorial Manager® and ProduXion Manager® from Aries Systems Corporation

Dear Editors,

I am pleased to submit our manuscript entitled “Accuracy of the ESCRS Toric IOL Power Calculator: Analysis of 6,753 Eyes” for your consideration.

In this study, we compared the predictive accuracy of four formulas integrated into the ESCRS calculator: Barrett, EVO, Hoffer® QST, and Kane. Notably, Barrett and EVO were evaluated using both measured and predicted posterior corneal curvature.

We confirm that this work is original, has not been published or submitted elsewhere, and has been approved by all authors.

Sincerely yours,

Li Wang

Synopsis

This retrospective study of 6,753 eyes compared four toric IOL formulas. The Kane formula demonstrated statistically superior accuracy, though clinical differences between formulas remained minimal

Accuracy of the ESCRS Toric IOL Power Calculator: Analysis of 6,753 Eyes

Short title: ESCRS IOL toric calculator

Janusz Skrzypecki, MD, PhD[1]; Douglas D. Koch, MD[2]; Li Wang, MD, PhD[2]

[1]Department of Experimental Physiology and Pathophysiology, Medical University of Warsaw, Warsaw, Poland

[2]Cullen Eye Institute, Department of Ophthalmology, Baylor College of Medicine, Houston, Texas

Financial Support: Supported in part by SRB Charitable Corp., Fort Worth, TX, the Sid W. Richardson Foundation, Fort Worth, TX, and an unrestricted grant from Research to Prevent Blindness, New York, NY (Koch and Wang).

Disclosures: J.Skrzypecki is a consultant for Johnson & Johnson Vision; D.D. Koch is a consultant for Alcon Laboratories, Inc., Carl Zeiss Meditec AG, and Johnson & Johnson Vision. L. Wang is a consultant for Carl Zeiss Meditec AG and Alcon Laboratories, Inc.

Corresponding author:

Li Wang

Department of Ophthalmology,

Cullen Eye Institute,

Baylor College of Medicine,

6565 Fannin, NC 205, Houston, Texas 77030, USA.

E-mail: liw@bcm.edu

ABSTRACT

Purpose: To compare the predictive accuracy of four toric IOL formulas (Barrett, EVO, Hoffer® QST, Kane) integrated in the ESCRS calculator. Barrett and EVO were tested with measured (Barrett M and EVO M) and predicted (Barrett P and EVO P) posterior corneal curvature.

Design: retrospective data analysis

Setting: multiple surgical centers in the USA

Methods: We analyzed 6753 eyes implanted with non-toric IOLs. Preoperative biometry, including posterior corneal curvature, was obtained using the IOLMaster 700. Predicted residual astigmatism was calculated via the proprietary websites of the four toric formulas integrated into the ESCRS IOL Power Calculator. Primary endpoints were the mean vector magnitude error and proportions of eyes with vector magnitude prediction errors ≤0.50 D.

Results: Kane demonstrated the lowest mean vector magnitude prediction error (0.537 D), significantly lower than EVO P (0.561 D), EVO M (0.57 D), Barrett M (0.579 D), Barrett P (0.583 D), and Hoffer®QST (0.622 D) (all $p<0.001$). This result was consistent in with-the-rule, against-the-rule, and oblique subgroups. Kane also achieved the highest proportion of eyes with prediction error ≤0.50 D ($p<0.001$). For all formulas, centroid values were significantly different from zero. The Kane formula showed the lowest total variances, significantly lower than all other formulas ($p<0.001$).

Conclusions: The Kane formula showed statistically superior predictive accuracy with the lowest errors and variances and the highest proportion of eyes within 0.50 D. However, differences among formulas were small, and their clinical relevance for toric IOL outcomes remains uncertain.

INTRODUCTION

Cataract surgery is increasingly regarded as a refractive procedure, with a strong emphasis on minimizing postoperative refractive error[1]. Up to 80% of patients achieve a residual spherical equivalent within ±0.5 diopters[1]. Nevertheless, up to 30-40% of eyes have pre-existing corneal astigmatism of 1 D or greater, which should be corrected during surgery to achieve a satisfactory visual outcome [2]. Several techniques are available, including incisions along the steep axis, paired opposite clear-corneal incisions, limbal relaxing incisions, or implantation of a toric intraocular lens (IOL), the latter being the most predictable technique[3].

Correcting preexisting corneal astigmatism poses unique challenges. Accurate and repeatable measurements of both anterior and posterior corneal curvature (PCC) are crucial; studies suggest that the repeatability of astigmatism measurements is only at the level of 0.4D[4]. Additionally, toric IOLs are often produced in 0.75 D increments, limiting the scope for fine-tuning[3]. IOL rotational stability is another important factor. Finally, it can be difficult to accurately predict surgeon-induced astigmatism, partly because keratometry readings themselves tend to have low repeatability[5, 6].

In recent years, methods for calculating toric IOL power have advanced substantially. One significant development was the Baylor Toric Nomogram, which accounts for the distinct impact of PCC on with-the-rule versus against-the-rule astigmatism[7]. Incorporating the Baylor Toric Nomogram into various formulas is suggested to improve predictions for residual cylinder[8]. Moreover, newer devices designed to measure PCC have enabled formulas that incorporate actual PCC measurements[5]. Nonetheless, studies comparing formulas that predict PCC with those that measure it directly have produced mixed results[9, 10].

The ESCRS toric calculator is a relatively new tool designed to streamline workflows by enabling direct comparisons of residual cylinder predictions from four modern toric formulas: Barrett Toric, EVO Toric, Hoffer®QST Toric, and the Kane Toric formula. These formulas have been compared previously in a small dataset, the present study expands on that

work by examining 6753 eyes implanted with non-toric IOLs[11]. Focusing on non-toric cases makes it possible to compare the formulas without the confounding effect of toric lens misalignment or rotation postoperatively, which is difficult to account for.

MATERIALS AND METHODS

We included 6753 eyes that were collected by Zeiss (Germany) from the Veracity Surgical Planner database. Surgeries were performed from May 2020 to October 2024 across 123 sites in United States. Only one eye per patient was analyzed. Preoperative biometry was performed with swept-source biometer IOL Master 700 (Zeiss, Germany). Only eyes with keratometry (K), total keratometry (TK), and posterior keratometry (PK) measurements of successful quality were included. None of the patients had undergone refractive surgery in the past or limbal relaxing incisions during the surgery. Postoperative follow-up took place at least three weeks after surgery, and only those eyes that achieved a postoperative manifest refraction and corrected distance visual acuity of 20/30 or better were included. Subjective refraction was converted to a 6-meter lane.

Since the current version of the ESCRS calculator does not support refractive predictions for EVO Toric and Barrett Toric IOLs using measured PCC, we accessed each calculator's proprietary website for this study. A Python script was used to input the data and retrieve the toric IOL predictions for monofocal lens implantation.

Statistical analysis followed the methodology described by Wang et al. and further refined by Koch et al[12, 13]. All calculations were performed in Excel (Microsoft, USA) and R Studio (Posit, USA) as per guidelines by Wang et al[12].

Statistical analysis was performed for the entire dataset as well as for subgroups based on astigmatism orientation with-the-rule (WTR), against-the-rule (ATR) as and oblique. Orientation was classified according to anterior corneal astigmatism measured with the IOLMaster 700: (1) WTR: steep meridian 67.5°–112.5°; (2) oblique: 22.5°–67.5° or 112.5°–

157.5°; (3) ATR: 0°–22.5° or 157.5°–180°. Furthermore, we also performed subgroup analyses of eyes implanted with the same IOL model.

For each formula, the prediction error of vector magnitude was calculated. Postoperative manifest astigmatism and the predicted residual astigmatism from each formula were first transformed into Cartesian coordinates (x- and y-components). The differences between the postoperative and preoperative components were then calculated. These differences were subsequently reconverted into polar notation, yielding a cylinder magnitude. The primary end-point of the study was the mean vector magnitude predictions error and percentage of eyes with vector magnitude prediction error ≤ 0.5D. The bootstrap-t method with Holm's correction for multiple comparisons was applied. To evaluate the distribution of vector magnitudes, a table was created showing the percentage of eyes falling within predefined error intervals: ≤0.50 D, ≤0.75 D, and ≤ 1.00 D. For each interval, the percentage of eyes was reported. Comparisons between formulas were performed using McNemar's Chi-squared test with continuity correction, appropriate for dependent data. Both nominal and Holm-adjusted p values were calculated for multiple comparisons.

To further assess vector direction and dispersion, we tested the null hypothesis that the bivariate mean and median of the whole dataset as well as dataset of subgroups do not differ from zero. Furthermore, total variances of x and y of vector components were compared between formulas[14-16]. Additionally, double-angle plots were included for each formula.

RESULTS

The characteristics of the study group are presented in Table 1. The Kane formula showed the lowest mean vector magnitude PE (0.537D), significantly lower than EVO P (0.561D), EVO M (0.57D), Barrett M (0.579D), Barrett P (0.583D), and Hoffer®QST (0.622D) (all $p < 0.001$, Table 2, Supplemental Table 1 and 2). Kane also achieved the highest proportion of eyes with a vector magnitude PE ≤ 0.50 D, 54%, with results significantly better than those

of the other formulas (all p<0.001; Table 3 and Supplemental Table 3 and 4). Furthermore, Kane demonstrated the smallest total variances of vector PEs across the entire dataset as well as within each subgroup (Supplemental Table 5)

Subgroup analyses by astigmatism orientation (ATR, WTR, and oblique) confirmed that Kane had the lowest mean vector magnitude PE (all p<0.001; Table 2, Supplemental Table 1 and 2) and the highest proportion of eyes with PE ≤ 0.50 D in ATR and oblique subgroups when compared to all formulas (all p<0.001, except for Kane vs. EVO P in oblique subgroup; p=0.03) and in WTR subgroups when compared with Barrett M, Barrett P and Hoffer®QST. (all p<0.001; Table 3 and Supplemental Table 3 and 4).

For all formulas, the bivariate mean and median of the vector PE were significantly different from zero, both in the overall dataset and in each astigmatism subgroup (Supplemental Table 6 and 7). In the whole group and subgroups of WTR, ATR, and oblique eyes, although the magnitudes of centroid values were small (0.040 D to 0.137 D), undercorrection was produced by Kane, EVO M, and Hoffer® QST, whereas overcorrection was produced by EVO P, Barrett M, and Barrett P (Table 2, Supplemental Table 6 and 7). Figure 1 shows the double-angle plots of the vector prediction errors by formula in the whole group.

A comparison within pairs of formulas that incorporate measured PCC and those that predict it showed the following results: in the overall study population, EVO P (0.561D) achieved a lower mean vector magnitude than EVO M (0.570D); p < 0.001. In contrast, Barrett M (0.579D) outperformed Barrett P (0.583D) (p = 0.02). In the ATR subgroup there was statistically significant difference between EVO P (0.625D) and EVO M (0.638D) (p<0.001) but not between Barrett P (0.646D) and Barrett M (0.646D). In the WTR subgroup there was significant difference between Barrett M (0.506D) and Barrett P (0.512D) (p=0.04), but not between EVO P (0.491D) and EVO M (0.494D). In the oblique subgroup, EVO P (0.546D) and Barrett M (0.564D) yielded lower mean values than EVO M (0.555D) and Barrett P (0.569D), respectively (p = 0.01 and p = 0.0495 respectively). Moreover, in the oblique subgroup, EVO P

provided a higher proportion of eyes with a prediction error ≤ 0.50 D compared with EVO M (p=0.017;52.5 % vs. 49.9 %). In other subgroups there were no significant differences in the percentage of eyes within prediction error ≤ 0.50 D between Barrett P and Barrett M as well as between EVO P and EVO M.

DISCUSSION

We found that the Kane formula demonstrated the highest accuracy in both scalar and vector analyses. The effect of incorporating PCC measurements varied by formula: it improved the performance of Barrett toric formula but not EVO toric formula.

Currently, two approaches are used to evaluate toric IOL outcomes: (1) analysis after toric IOL implantation, and (2) analysis after monofocal IOL implantation. While the monofocal approach (used in this study) may seem less reflective of real-world toric outcomes, it avoids confounding from postoperative IOL alignment and compensations for clinically irrelevant IOL rotation[10, 11]. Moreover, it facilitates the collection of larger datasets. To our knowledge, this is the largest study to date assessing the precision of modern toric IOL calculators.

Our results contrast with those of Liu et al., who reported the EVO formula as the most accurate and Kane as the least accurate among the toric formulas integrated into the ESCRS calculator. However, their dataset included only 85 eyes, and the analysis was performed after toric IOL implantation, introducing potential bias from suboptimal IOL positioning. Furthermore, the authors did not report results for EVO M formula[11]. Similarly, Yang et al., in a study of 146 eyes after toric IOL implantation, compared Barrett, EVO, Hoffer®QST, and Abulafia-Koch combined with Hill-RBF, and found EVO to be the most accurate, while Barrett with estimated PCC measurements was the least accurate. However, in this study authors did not measure PCC directly[17].

Stewart et al., analyzing 418 eyes with monofocal (non-toric) IOLs and following

formulas: Barrett P, Barrett M, EVO P, EVO M and Holladay I with Abulafia-Koch regression, reported results consistent with ours: EVO P outperformed Barrett P and Barrett M[18].

The role of PCC measurement in toric IOL calculations remains of particular interest to surgeons. Prior studies have yielded mixed results. A small study by Skrzypecki et al. found no significant difference between the Barrett toric calculator with and without PCC measurement, whereas Wang et al. reported a significant reduction in mean vector magnitude PE with measured PCC[9, 10]. Nevertheless, the result did not appear clinically relevant. Stewart et al. suggested that PCC measurement may be especially beneficial in eyes with non-vertical posterior corneal astigmatism. Other smaller studies also indicate that it may be useful in select cases, such as eyes with unusually steep posterior corneal surfaces[19].

In our cohort, results support prior observations that the mean vector magnitude PE tends to hover around 0.5 D, with only ~50% of eyes achieving PE ≤ 0.5 D. Although inter-formula differences reached statistical significance, they were not clinically meaningful, regardless of astigmatism orientation. This likely reflects the limitations of current corneal measurement technologies, where precision and repeatability may constrain outcomes more than the formulas themselves, as well as design of toric IOL, which are usually manufactured in 0.5 D or 0.75 D steps.

One key limitation of our analysis is its retrospective nature. We relied on pre-existing, anonymized clinical data, meaning there was no opportunity to verify crucial aspects of each case or standardize data collection across centers. For example, our dataset included an eye with a average K value as low as 36.40 D, but we could not confirm whether this patient had undergone prior corneal refractive surgery; such unrecorded procedures could significantly influence postoperative outcomes. Likewise, the manifest refraction data were performed by different clinicians across multiple clinics, and we had no way to identify or evaluate individual techniques and inter-observer variability. These factors collectively introduce potential sources of bias and heterogeneity that should be considered when interpreting our findings

In conclusion, this is the largest study to date comparing modern toric IOL formulas with and without PCC measurement. In our dataset of 6753 eyes, both scalar and vector analyses identified the Kane formula as statistically the most accurate. However, the differences between formulas do not appear to be clinically significant.

What was known

- The ESCRS toric calculator incorporates four contemporary formulas.
- Previous studies have reported mixed results on whether incorporating posterior corneal curvature improves the accuracy of toric IOL power prediction.

What this paper adds

- This is the largest study to date comparing contemporary toric IOL formulas.
- The Kane formula, which does not incorporate posterior corneal curvature measurements, demonstrated the highest accuracy.
- While some formulas showed statistically significant differences, these were clinically small; improved outcomes in astigmatism correction might require advances in IOL design and measurement precision.

REFERENCE

1 Melles RB, Holladay JT, Chang WJ. Accuracy of intraocular lens calculation formulas. Ophthalmology 2018; 125: 169-178

2 Joshi RS, Jadhav SA. Frequency of corneal astigmatism in patients presenting for senile cataract surgery at a teaching hospital in indian rural population. Asia Pac J Ophthalmol (Phila) 2020; 9: 126-129

3 Al-Mohtaseb Z, Steigleman WA, Pantanelli SM, Lin CC, Hatch KM, Rose-Nussbaumer JR, Santhiago MR, Olsen TW, Kim SJ, Schallhorn JM. Toric monofocal intraocular lenses for the correction of astigmatism during cataract surgery: a report by the american academy of ophthalmology. Ophthalmology 2024; 131: 383-392

4 Kanellopoulos AJ, Asimellis G. Distribution and repeatability of corneal astigmatism measurements (magnitude and axis) evaluated with color light emitting diode reflection topography. Cornea 2015; 34: 937-944

5 Hirnschall N, Findl O, Bayer N, Leisser C, Norrby S, Zimper E, Hoffmann P. Sources of error in toric intraocular lens power calculation. J Refract Surg 2020; 36: 646-652

6 Naeser K. Assessment and statistics of surgically induced astigmatism. Acta Ophthalmol 2008; 86: 349

7 Koch DD, Jenkins RB, Weikert MP, Yeu E, Wang L. Correcting astigmatism with toric intraocular lenses: effect of posterior corneal astigmatism. J Cataract Refract Surg 2013; 39: 1803-1809

8 Reitblat O, Levy A, Kleinmann G, Abulafia A, Assia EI. Effect of posterior corneal astigmatism on power calculation and alignment of toric intraocular lenses: Comparison of methodologies. J Cataract Refract Surg 2016; 42: 217-225

9 Skrzypecki J, Sanghvi Patel M, Suh LH. Performance of the Barrett Toric Calculator with and without measurements of posterior corneal curvature. Eye (Lond) 2019; 33: 1762-1767

10 Wang L, Koch DD. Comparison of accuracy of a toric calculator with predicted vs measured posterior corneal astigmatism. J Cataract Refract Surg 2023; 49: 29-33

11 Liu C, Wang M, Long D, Zhang Y, Chen Y, Wu Q. Comparison of the Accuracy of toric intraocular lens formulas used by the online calculator of the european society of cataract and refractive surgeons. J Refract Surg 2025; 41: e120-e130

12 Wang L, Holladay JT, Wilcox RR, Koch DD. Wrap-up R functions for statistical analysis of spherical equivalent and astigmatic prediction errors. J Cataract Refract Surg 2023; 49: 890-891

13 Koch DD, Holladay JT, Naeser K, Abulafia A, Wang L, Stulting RD, Kohnen T. Standards for Analyzing Astigmatic Outcomes II: Recommended statistical methods. J Cataract Refract Surg 2025

14 Holladay JT, Wilcox RR, Koch DD, Wang L. Statistics of prediction error for non-Gaussian dependent and independent datasets. J Cataract Refract Surg 2023; 49: 440-442

15 Holladay JT, Wilcox RR, Koch DD, Wang L. Contemporary statistical analysis of refractive error in adults, pseudophakia, and refractive surgery. J Cataract Refract Surg 2022; 48: 1346

16 Holladay JT, Wilcox RR, Koch DD, Wang L. Astigmatism analysis and reporting of surgically induced astigmatism and prediction error. J Cataract Refract Surg 2022; 48: 799-812

17 Yang S, Han H, Lee HS. Comparative accuracy of five modern toric intraocular lens formulas. Am J Ophthalmol 2025; 274: 1-8

18 Stewart S, Yeo TK, Moutari S, McNeely R, Moore JE. Accuracy of toric intraocular lens formulas with measured posterior corneal astigmatism of different orientations. Am J Ophthalmol 2024; 266: 26-36

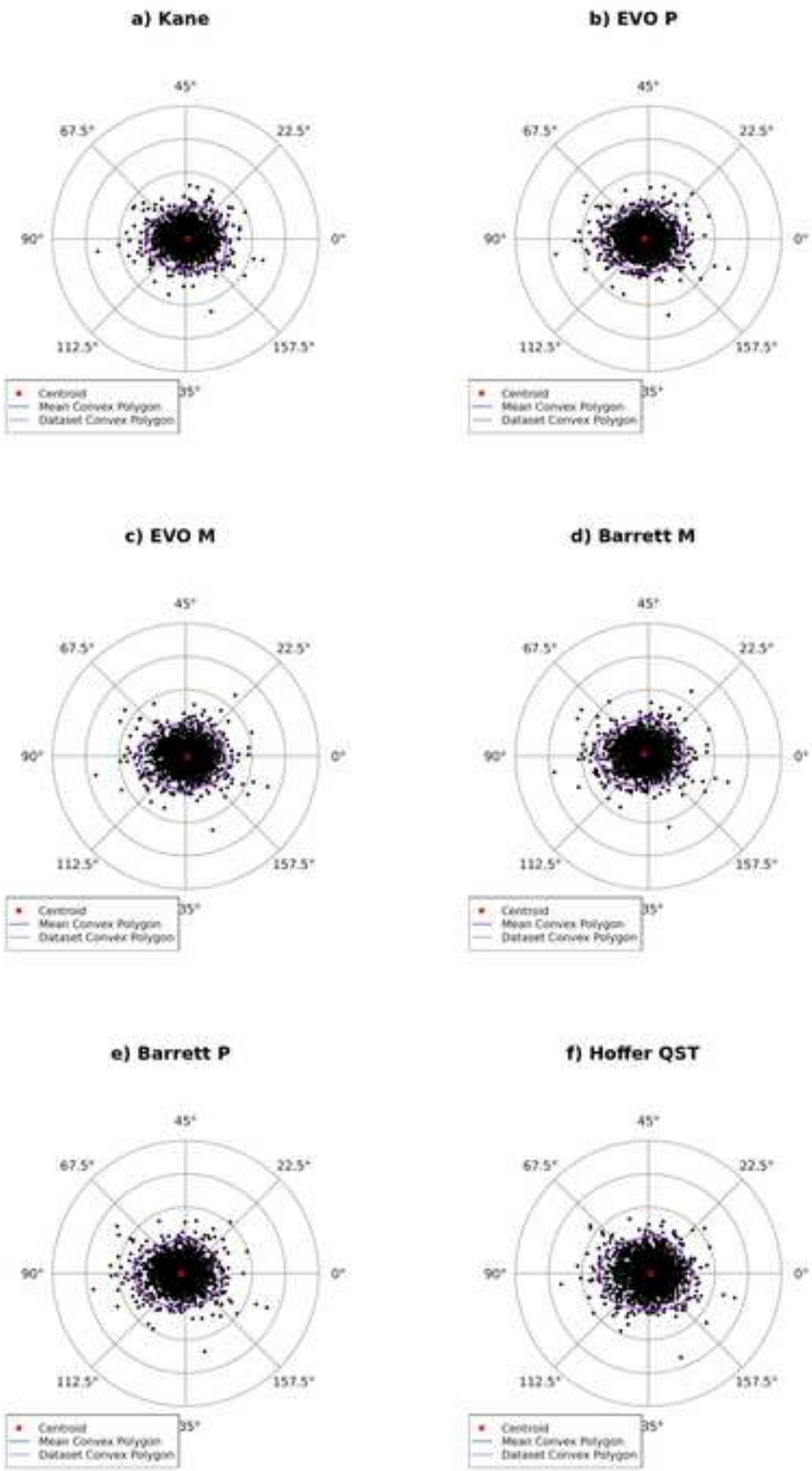
19 Reitblat O, Levy A, Megiddo Barnir E, Assia EI, Kleinmann G. Toric IOL calculation in eyes with high posterior corneal astigmatism. J Refract Surg 2020; 36: 820-825

FIGURE LEGEND:

Figure 1. Double-angle plots of the vector prediction error by formula.

Figure

Click here to access/download;Figure;Figure 1 .jpg



Table

Table 1. General characteristics of the group

Feature	Mean ± SD	Range
No. female	3889	
Age (years)	71.7 ± 8.5	22–97
Axial length (mm)	23.99 ± 1.23	19.75–31.07
Average Ks (D)	44.02 ± 1.57	36.40 – 52.53
K1 (D)	43.49 ± 1.60	36.27–51.26
K2 (D)	44.66 ± 1.63	36.89–53.41
K Astigmatism (D)	1.11 ± 0.6	0.50-7.76
Average PK (D)	-5.86 ± 0.25	-4.71 – -7.28
PK1 (D)	-5.74 ± 0.24	-4.68– -6.93
PK2 (D)	-5.99 ± 0.27	-4.74– -7.63
WTW (mm)	12.00 ± 0.42	10.19–13.84
CCT (µm)	551.03 ± 36.28	406.01–705.47
ACD (mm)	3.17 ± 0.39	1.57–4.74
LT (mm)	4.64 ± 0.43	2.79–6.24
IOL power (D)	19.92 ± 3.48	0.00–34.00

K-keratometry; PK – posterior keratometry; WTW – white-to-white; CCT- central corneal thickness; ACD – anterior chamber depth; LT – lens thickness; IOL – intraocular lens

Table 2. Mean vector magnitude of astigmatism prediction error (D) and centroid values (D @ degrees).

	Kane	EVO P	EVO M	Barrett M	Barrett P	Hoffer® QST
All (n=6753)						
Mean	0.537	0.561	0.57	0.579	0.583	0.622
Centriod	0.065 @ 5	0.106 @ 89	0.045 @ 170	0.131 @ 77	0.119 @ 91	0.064 @ 1
ATR (n=2663)						
Mean	0.595	0.625	0.638	0.646	0.646	0.682
Centroid	0.074 @ 2	0.098 @ 91	0.050 @ 170	0.124 @ 77	0.113 @ 92	0.070 @ 178
WTR (n=1992)						
Mean	0.472	0.491	0.494	0.506	0.512	0.55
Centroid	0.065 @ 9	0.108 @ 87	0.040 @ 173	0.137 @ 76	0.121 @ 89	0.064 @ 6
Oblique (n=2098)						
Mean	0.524	0.546	0.555	0.564	0.569	0.612
Centroid	0.054 @ 6	0.114 @ 90	0.044 @ 167	0.134 @ 78	0.126 @ 92	0.059 @ 179

ATR – agains-the-rule; WTR- with-the-rule

Table 3. Proportion of eyes (%) with vector prediction error ≤ 0.50 D, ≤ 0.75 D, and ≤ 1.00 D

	Kane	EVO P	EVO M	Barrett M	Barrett P	Hoffer® QST
All						
0.50 D	54	52	50	50	49	43
0.75 D	78	76	75	74	74	70
1.00 D	91	89	89	88	88	86
ATR						
0.50 D	47	44	43	42	42	36
0.75 D	72	70	68	66	67	63
1.00 D	88	85	85	84	83	81
WTR						
0.50 D	63	61	61	59	59	51
0.75 D	84	83	83	82	81	78
1.00 D	94	93	92	92	92	91
Oblique						
0.50 D	55	52	50	50	50	44
0.75 D	80	78	78	76	76	70
1.00 D	92	90	91	89	89	87

ATR – agains-the-rule; WTR- with-the-rule

Click here to access/download
Supplementary Material
Supplemental Table 1 final .xlsx

Click here to access/download
Supplementary Material
Supplemental Table 2 final.xlsx

Click here to access/download
Supplementary Material
Supplemental Table 3 final .xlsx

Click here to access/download
Supplementary Material
Supplemental Table 4 final .xlsx

Click here to access/download
Supplementary Material
Supplemental Table 5 final.xlsx

Click here to access/download
Supplementary Material
Supplemental Table 6.xlsx

Click here to access/download
Supplementary Material
Supplemental Table 7.xlsx